

HYVINKAA HYVINKAANKYLA, Finland

WMO: 028290

Lat: 60.5958N

Lon: 24.8031E

Elev: 88

StdP: 100.28

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -23.1	(c) -19.6	(d) -25.5	(e) 0.4	(f) -22.8	(g) -22.0	(h) 0.5	(i) -19.3	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 10.3	(c) 27.7	(d) 19.0	(e) 25.7	(f) 17.7	(g) 23.9	(h) 16.9	(i) 20.5	(j) 25.4	(k) 19.2	(l) 23.9	(m) 17.9	(n) 22.3	(o) N/A	(p) N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.7	22.9	17.4	12.6	21.3	16.1	11.6	20.0	59.4	25.6	55.0	23.9	50.9	22.3	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) N/A	(b) N/A	(c) N/A	(e) -25.3	(f) 29.1	(g) 5.2	(h) 2.5	(i) -29.0	(j) 30.9	(k) -32.0	(l) 32.4	(m) -34.9	(n) 33.8	(o) -38.7	(p) 35.7		
(4) N/A	(5) N/A	(6) N/A	(e) -25.5	(f) 21.4	(g) 5.1	(h) 1.6	(i) -29.2	(j) 22.6	(k) -32.2	(l) 23.6	(m) -35.1	(n) 24.5	(o) -38.8	(p) 25.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.2	-6.8	-5.0	-1.9	4.0	10.4	14.2	17.2	15.4	11.0	4.6	1.2	-2.7
	DBStd	9.12	6.47	6.19	4.58	3.43	4.17	3.47	3.19	2.83	3.24	3.60	4.25	5.42
	HDD10.0	2394	521	419	369	181	45	6	0	1	26	169	263	393
	HDD18.3	4833	779	653	627	428	246	131	61	99	219	425	513	652
	CDD10.0	643	0	0	0	2	58	131	223	168	57	3	0	0
	CDD18.3	41	0	0	0	0	1	7	26	7	0	0	0	0
	CDH23.3	492	0	0	0	0	36	99	276	81	0	0	0	0
	CDH26.7	91	0	0	0	0	3	13	59	16	0	0	0	0
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	PrecAvg	623	51	38	33	33	38	63	71	71	55	63	59	47
(15) Precipitation	PrecMax	799	91	93	68	70	78	117	226	145	134	161	163	117
	PrecMin	478	8	1	6	5	3	15	1	10	16	14	4	6
	PrecStd	84	20	22	15	17	20	27	46	31	32	33	32	23
	PrecStd	84	20	22	15	17	20	27	46	31	32	33	32	23
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.1	5.8	10.4	19.9	27.1	28.5	30.6	28.8	22.0	14.1	10.3	8.3
		MCWB	3.3	2.9	5.2	10.9	15.7	18.8	20.5	19.2	16.3	11.7	9.3	7.5
	2%	DB	2.9	3.6	7.3	16.9	24.1	26.0	28.5	25.4	19.6	12.4	8.9	6.0
		MCWB	2.4	2.4	3.7	8.8	14.9	16.9	19.9	18.3	15.3	10.8	8.5	5.3
	5%	DB	2.0	2.4	5.3	13.9	21.8	24.1	26.6	23.3	17.9	11.4	7.5	4.5
		MCWB	1.7	1.8	2.7	7.9	13.6	16.2	19.0	17.1	14.0	10.2	7.0	3.9
	10%	DB	0.8	1.6	3.5	11.2	19.4	21.9	24.5	21.5	16.6	10.4	6.5	3.2
		MCWB	0.5	1.2	1.7	6.0	12.2	15.0	18.1	16.1	13.8	9.3	6.1	2.7
	0.4%	WB	3.5	3.6	5.7	11.6	17.7	20.2	22.2	21.9	17.2	12.7	9.6	7.7
		MCDB	3.8	4.7	8.4	18.3	23.4	25.8	27.7	26.4	20.4	13.4	10.0	8.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	2.6	2.6	4.2	9.8	16.0	18.6	21.1	19.4	16.0	11.5	8.4	5.4
		MCDB	3.0	3.4	6.4	15.4	21.0	24.5	26.3	23.3	18.4	12.1	9.0	6.0
	5%	WB	1.7	1.9	3.1	8.2	14.5	17.0	20.0	18.0	15.2	10.5	7.1	3.9
		MCDB	2.0	2.4	4.8	12.8	20.3	22.1	25.0	21.7	17.2	11.1	7.4	4.3
	10%	WB	0.6	1.2	2.1	6.8	13.0	15.7	18.8	17.0	14.2	9.3	6.1	2.8
		MCDB	0.8	1.5	3.2	10.3	18.3	20.7	23.4	20.2	15.8	10.1	6.4	3.1
(35) Mean Daily Temperature Range	5% DB	MDBR	5.6	5.6	7.9	9.9	11.7	10.5	10.3	10.1	8.3	6.4	3.9	4.3
		MCDBR	3.7	4.8	9.4	15.7	16.1	14.5	13.4	13.5	10.8	8.0	4.6	4.8
		MCWBR	3.7	4.1	6.2	8.8	7.9	6.9	6.1	7.0	6.9	6.3	4.5	4.6
		MCDBR	3.7	4.0	8.0	13.9	13.7	13.0	11.8	11.1	9.3	6.2	4.6	4.6
	5% WB	MCWBR	3.7	3.5	5.8	8.2	7.2	6.7	5.9	6.3	6.3	5.6	4.5	4.4
		MCDBR	3.7	3.5	5.8	8.2	7.2	6.7	5.9	6.3	6.3	5.6	4.5	4.4
	Clear-Sky Solar Irradiance	taub	0.268	0.304	0.329	0.358	0.355	0.352	0.374	0.365	0.339	0.327	0.294	0.379
		taud	2.122	2.185	2.256	2.368	2.428	2.461	2.419	2.441	2.509	2.403	2.209	2.622
(44) All-Sky Solar Radiation	Ebn at Noon	470	672	789	839	871	878	846	825	788	656	436	326	27
		Edh at Noon	41	71	92	100	101	100	102	92	73	58	39	27
	RadAvg	0.24	0.81	2.29	3.82	5.14	5.35	5.18	3.99	2.42	1.06	0.30	0.14	0.02
		RadStd	0.04	0.17	0.30	0.39	0.56	0.43	0.48	0.42	0.32	0.18	0.06	0.02

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	-141	N/A	N/A	N/A

Nomenclature: See separate page